Thermodynamics of Angular Propulsion in Fluids

Jeliazko G. Polihronov and Anthony G. Straatman
The Department of Mechanical and Materials Engineering, Western University, London, Ontario, Canada N6A 5B9
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The presented study examines the energetics of confined fluid flow in a rotating reference frame. Parallels are drawn to the corresponding scenario of rectilinear motion, where ejection of fluid results in linear propulsion of the frame. Absorption of flow energy into the frame motion leads to cooling of the ejected fluid. Relevance of the observed energetics to the temperature separation phenomenon in Ranque-Hilsch vortex tubes is discussed.

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Introduction.—This work revisits the concept of momentum conservation and the corresponding propulsion imparted to a reference frame by an ejected fluid. The focus is on constrained flows within moving frames, where flow confinement results in a well-defined physical problem. The energetics of the phenomena are examined with a particular goal in mind—namely, to predict the fluid temperature as observed in different frames of reference as well as describe the underlying physics leading to such observations. Attention is devoted to the applicability of the presented physical model to rotational flows, which exhibit radial temperature separation. A most relevant example is the vortex tube effect, discovered in 1933 by the French physicist Georges J. Ranque [1]. The effect has now been studied for nearly 80 years, yet while a number of models have been proposed, they remain a subject of debate [2]. The fundamental reason for this is the complexity of vortex tube flow obscuring the underlying physics, which in its turn obfuscates any concise understanding of the effect. Notwithstanding, interest in the vortex tube phenomena remains high, as demonstrated by a present day literature search in the Google Scholar database resulting in 4240 references to published documents discussing the topic of vortex tube airflow.

In its approach to the problem, the present work assumes *a priori* that a rotating flow can be discretized and examines the behavior of its discrete element—a paradigm, through which the long-standing physical phenomenon of temperature separation unravels and becomes accessible to analysis. The main reasoning in this work, therefore, follows along the lines of establishing relative contexts of a stationary and moving observer, positioned in their corresponding reference frames, followed by an examination of relative flow motion and the relevant conservation laws.

In the physics of fluids, the thermodynamic (or static) temperature T_s is that which corresponds to thermal equilibrium and is the same in all frames of reference. The total, or stagnation, temperature is an effective temperature that originates from the total (or stagnation) enthalpy $h = h_s + \mathbf{v}^2/2$ via division by the isobaric heat capacity c_p, and takes the form

$$T \equiv T_s + \frac{\mathbf{v}^2}{2c_p}, \tag{1}$$

where $\mathbf{v}$ is the fluid velocity. Because the total temperature contains $\mathbf{v}$, it is, consequently, frame-dependent. In a moving frame F', this temperature becomes

$$T_{\mathrm{rel}} \equiv T_s + \frac{\mathbf{v}'^2}{2c_p}, \tag{2}$$

where $\mathbf{v}'$ is the flow velocity relative to the frame. In adiabatic duct flow, the conservation of energy demands that the total enthalpy is conserved. Thus, utilizing the connection between total enthalpy and total temperature, energy conservation can also be expressed as

$$T = \text{const.} \tag{3}$$

under adiabatic flow conditions [3].

Angular propulsion.—Consider the reference frame F', rotating about the z-axis with constant angular velocity $\boldsymbol{\omega} = \text{const}$. Energy conservation in rotating fluid flows has the form [4]

$$T_s + \frac{\mathbf{v}'^2 - (\boldsymbol{\omega} \times \mathbf{r})^2}{2c_p} = \text{const} \tag{4}$$

under adiabatic conditions. Let the rotating frame F' be attached to a fluid flow system, comprising a tank of compressible fluid under high pressure and room temperature T_∞, connected to the inlet of an adiabatic duct, as shown in Figure (1). The compressed fluid is allowed to flow through the duct where it gradually expands, accelerates and exits at the center of the frame. The velocity addition formula for the system is

$$\mathbf{v} = \mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}. \tag{5}$$

Expressing $\mathbf{v}'$ and substituting it into the energy conservation condition (4) yields

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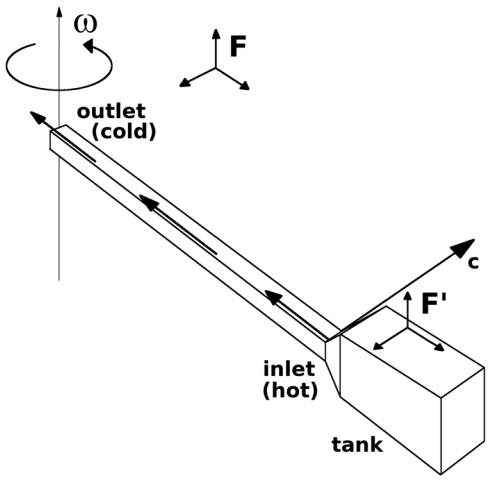


FIG. 1. Rotating duct.

$$T_s + \frac{\mathbf{v}^2 - 2\mathbf{v}\cdot\boldsymbol{\omega}\times\mathbf{r} + (\boldsymbol{\omega}\times\mathbf{r})^2 - (\boldsymbol{\omega}\times\mathbf{r})^2}{2c_p} = \text{const.} \tag{6}$$

Reworking this expression to include the total fluid temperature (1) seen in the stationary frame yields

$$T - \frac{\mathbf{v}\cdot\boldsymbol{\omega}\times\mathbf{r}}{c_p} = \text{const.} \tag{7}$$

Therefore, the observer in the stationary frame F will report a temperature difference

$$\Delta T = T(\text{inlet}) - T(\text{outlet}) = \frac{\mathbf{v}_{\text{inlet}}\cdot\boldsymbol{\omega}\times\mathbf{r}_{\text{inlet}}}{c_p} - \frac{\mathbf{v}_{\text{outlet}}\cdot\boldsymbol{\omega}\times\mathbf{r}_{\text{outlet}}}{c_p} \tag{8}$$

between the high-energy peripheral flow and the low-energy flow at the rotation center. Since in this particular fluid flow system the duct is straight, $\mathbf{v}' \perp \boldsymbol{\omega}\times\mathbf{r}$ everywhere, and because the flow exits at the rotation center, $\mathbf{r}_{\text{outlet}} = 0$. If we denote the peripheral tip speed of the duct $\boldsymbol{\omega}\times\mathbf{r}_{\text{inlet}}$ as $\mathbf{c}$, then (8) reduces to

$$\Delta T = T(\text{inlet}) - T(\text{outlet}) = \frac{\mathbf{c}^2}{c_p}. \tag{9}$$

Thermodynamics of the flow is interpreted in F and F' as follows: *According to an observer in the moving frame F'.*—(i). Both static and relative total temperatures in the fluid tank are equal to T_∞. (ii). The tank fluid expands through the duct and does work to overcome the centrifugal gravitational potential $-(\boldsymbol{\omega}\times\mathbf{r})^2/2$; the exiting fluid has lost internal energy and gained gravitational potential energy. (iii). The fluid accelerates through the duct, due to expansion, and experiences the deflecting action of the Coriolis force. (iv). At the outlet, the exiting fluid has a higher velocity than at the duct inlet due to expansion, but has lost internal energy and is $\frac{\mathbf{c}^2}{2c_p}$ cooler than T_∞.

According to an observer in the stationary frame F.—(i). The total temperature in the pressurized fluid tank is $T = T_\infty + \frac{\mathbf{c}^2}{2c_p}$ due to the motion of F'. (ii). The fluid speed at the duct inlet is equal to $\mathbf{c}$ and the temperature is equal to the temperature in the fluid tank $T(\text{inlet}) = T_\infty + \frac{\mathbf{c}^2}{2c_p}$. (iii). High-energy fluid decelerates as it approaches the outlet; that is, while the radial velocity increases, the tangential velocity goes to zero, resulting in a substantial net deceleration. (iv). At the outlet, the exiting fluid has low velocity and has also lost internal energy. This conclusion contradicts the intuition of the stationary observer, since a high-energy volume of compressible fluid is expected to exhibit a static temperature rise when brought to rest adiabatically.

A parallel can be drawn between the rotating system described above and its corresponding scenario of flow in a reference frame moving in a straight line with velocity $\mathbf{c}$. In a stationary frame of reference, the tank attached to the leading end of the moving duct will appear heated to $T_\infty + \mathbf{c}^2/2c_p$ due to the motion of the frame. In the moving frame of reference, for compressible fluids, energy conservation (3) requires a flow velocity increase and a corresponding static temperature decrease towards the duct outlet as seen in (2). If we choose conditions such that the magnitude of the relative velocity with which the fluid exits the moving frame is equal to $\mathbf{c}$, the static temperature at the outlet is $T_s(\text{outlet}) = T_\infty - \mathbf{c}^2/2c_p$. Then, the stationary observer will report a temperature difference between the ends of the duct equal to $\Delta T = T(\text{inlet}) - T(\text{outlet}) = \mathbf{c}^2/c_p$, an expression identical to (9).

When comparing the cases of rectilinear and rotational motion, the accounts of the moving observers will differ since the laws of physics are not invariant with respect to change from inertial into noninertial frame. However, the accounts of the stationary observers will be identical, recording identical energy losses in the fluid expelled from the moving frame. In the familiar case of rectilinear motion, it is a known fact that the energy losses in the fluid are due to the propulsion imparted to the moving frame. In the case of rotation, the inward motion of mass increases the rotational energy of the frame; this motion is caused by the fluid producing work via adiabatic expansion. Energy and angular momentum conservation demand that energy loss due to a change in radial position, as well as the amount of work causing this change, be imparted to the system [5]. Therefore, the imposing conclusion is that while linear propulsion is due to conservation of linear momentum, conservation of angular momentum leads to angular propulsion in the case of rotation.

Temperature separation.—It is seen that the energy conservation condition (7) imposes radial dependence in the total temperature known as temperature separation. It is a physical phenomenon, in which rotating fluid flow appears heated at the periphery and cooled at the center of rotation.

In the case of rectilinear motion, it was shown that cooling of the ejected fluid is due to rectilinear propulsion imparted to the moving frame. Therefore, in the case of rotation, cooling of the ejected fluid is due to conservation of angular momentum and the corresponding angular propulsion imparted to the rotating frame. It is this critical element that leads to a clear understanding of the temperature separation effect in fluids. Since the energy conservation requirement (4) applies under adiabatic conditions, it prohibits heat exchange through the duct walls in the system in Fig. 1. Therefore the cooling of the fluid (9) is a result of adiabatic expansion, during which the fluid does work on its surroundings by propelling the moving reference frame.

What are the conditions under which no cooling is observed? If the reference frame containing the flow is not moving, no cooling will be observed since the frame is unable to absorb the flow energy. For a duct at rest, where the exiting flow velocity has been chosen to be equal to $\mathbf{c}$, no temperature decrease is observed. Cooling in the stationary frame is produced only when the duct system is moving and able to absorb the energy of the flow as thrust or propulsion. The produced temperature separation ΔT grows with the magnitude of the frame velocity $\mathbf{c}$ and is limited by the speed of sound in the surrounding fluid for practical reasons. ΔT is always symmetric with respect to the ambient temperature T_∞ and equal to $\mathbf{c}^2/c_p$. When $\mathbf{c}$ is nearly equal to the speed of sound at sea level (340 m/s), $\Delta T = 115.2$ K. The heating of $\mathbf{c}^2/2c_p = 57.6$ K is entirely dynamic and due to the motion of the duct periphery with velocity $\mathbf{c}$; the cooling is due to an adiabatic expansion needed to overcome the centrifugal potential barrier and has magnitude of $\mathbf{c}^2/2c_p = 57.6$ K.

It is also important to note that compressibility of the fluid is vital for storing internal energy, which would later be imparted to the frame upon decompression as well as result in a reduction of static temperature. In the case of incompressible fluids, energy is still transferred to the frame due to angular momentum conservation; however, this cannot produce cooling as the fluid is unable to give up internal energy. The same conclusion is found in the work of Balmer [6], where water was used as the working fluid in a vortex tube. Cooling was not achieved in any of the conducted experiments in [6] and fluid at the periphery was reported to have an elevated temperature. This result is consistent with angular propulsion imparted on the rotating fluid, resulting in high kinetic energies at the periphery consequently leading to heating through friction.

It is also worth noting that the magnitude of ΔT does not depend on the radial size of the rotating system, as long as its peripheral velocity is equal to $\mathbf{c}$ in the stationary frame. In addition, centrifugal and Coriolis forces alone cannot alter the total temperature of the flow, since no work is subtracted from the fluid under gravity.

Flow through the rotating duct shown in Fig. 1 was also computed using the commercial computational fluid dynamics (CFD) solver FLUENT [7] to demonstrate that the results of the presented theoretical model are also obtained by discretely solving the differential transport equations for mass, momentum and energy. Simulations were performed with air as an ideal gas using the three-dimensional, double precision discretization model for compressible flow. The standard version of the $k-\epsilon$ model with wall-functions was used to characterize turbulence effects, and the second-order upwind discretization scheme was used to model advection in the transport equations. Since physical scale is not a factor in the current treatment, the duct was given a length of 15 m and rectangular cross-sectional dimensions 0.3 m $\times$ 0.4 m with no-slip, adiabatic walls. Smaller or larger ducts will produce the same effect provided the rotational speed is adjusted to develop the same pressure gradient across the duct. The computational grid was refined near the duct walls to ensure proper approximation by the turbulent wall functions. In all calculations, a mass flow rate of 3 kg/s with a low turbulence level was imposed at the duct inlet; a pressure level was set at the duct outlet with zero-gradient conditions on all other variables. Specification of a rotating reference frame was used to impart motion on the duct and all effects of centrifugal acceleration were included; the highest rotational speed was selected such that the peripheral velocity of the duct $\mathbf{c}$ remained subsonic. Energy, momentum and mass conservation were reached in all simulations, with residuals decreasing smoothly to below 10^{-13}. Table I compares the theoretical $\Delta T = \omega^2 r^2/c_p$ ($r = 15$ m) with its corresponding total temperature difference predicted by FLUENT for different rotational speeds. The CFD predictions approximate the theoretical result to within 1.5% in all cases. This comparison shows that the numerical values for ΔT given by Eq. (9) are also obtained using another well-established method; it should be bourne in mind that CFD utilizes discretization and turbulence modeling and as such represents an approximation to the physical phenomena described above.

Relevance to the vortex tube effect.—While the setup in Figure 1 is not identical to a vortex tube, it demonstrates the essential physical characteristics of the vortex tube flow, namely, spiral flow geometry accompanied by radial pressure and temperature behavior. Therefore, a rotating duct can be considered a discrete element of the vortex tube flow field. It presents a simplification in the description of vortex tube flow, which allows for a succinct explanation of the vortex tube phenomenon. For the rotating duct, flow is driven from the periphery to the center by a pressure gradient that opposes the centrifugal gravitational field

TABLE I. ΔT for different rotation rates.

ω, [rad/s]	0	2	5	10	15	20
ΔT, CFD [K]	0	0.89	5.53	22.08	49.68	88.30
ΔT, Eq. (9) [K]	0	0.90	5.61	22.42	50.45	89.69

induced by rotation. Energy is imparted by the expanding fluid to propel the rotating frame via the interface between the fluid and the solid (i.e. the duct wall). In this manner, maximum energy exchange occurs and the maximum possible temperature separation is observed. In the case of a vortex tube, flow is driven from the periphery of the tube to the center by a pressure gradient that opposes the induced gravitational field, but the expanding fluid can only transfer energy to the rotating frame (the fluid itself) via fluid friction, leading to less efficient cooling than that for the confined flow.

A key difference between the rotating duct and the vortex tube is the necessity of a hot fluid outlet in the latter. The hot outlet is not required in the rotating duct because the compressed fluid source is rotating with the duct; the only heating that occurs is due to fluid friction opposing the flow towards the duct outlet. In a vortex tube, the fluid enters the tube at the periphery to generate the swirling flow, and to set up the (centrifugal) gravitational field and the pressure gradient. Because of the high flow speeds required to set up the required gravitational field, fluid friction results in significant viscous dissipation at the periphery, which must be removed to achieve any cooling effect at the cold outlet (relative to the inlet). If the hot outlet were closed, the fluid leaving the system would simply absorb all of the viscous heat and leave the system warmer than it entered.

In terms of the magnitude of temperature separation, the control parameters in either case are the rotational speed of the fluid and the radius from the center to the periphery, since this sets up the strength of the centrifugal gravitational field, which dictates the pressure gradient from the periphery to the center. This pressure gradient dictates the maximum temperature drop that can be achieved by expansion of the fluid as it flows towards the cold outlet.

Conclusion.—When radial flow of a compressible fluid takes place in a uniformly rotating adiabatic duct, the resulting cooling that is observed at the center of rotation is due to adiabatic expansion of the fluid as well as conservation of angular momentum, which demands transfer of internal and rotational energy of the moving mass to the rotational energy of the system. Cooling cannot be produced in a stationary duct by gravity, as frame motion is required for an energy transfer to occur. Compressibility is another required factor for cooling since it reflects the ability of the fluid to give away internal energy. Of key importance, is that the confined rotating fluid flow system presented in this work exhibits the essential physics of the vortex tube flow, namely, radial temperature and pressure gradients as well as velocity fields and flow geometry. It is therefore plausible to consider this simplified flow system as a discrete element of vortex tube flow, which provides a concise understanding of the observed temperature separation phenomenon.

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